Question 1

Fresh B, A, Y, SA, C and G are needed for each candidate. More of the solutions should be available if requested by candidates. Solutions and reagents provided to the candidates should be supplied in a suitable beaker, or container, for removal of the solution using a syringe.

B, at least 50 cm^3 of 0.02% bromothymol blue solution, in a covered beaker or container, labelled **B**.

This is prepared by making a 2.0% stock solution by dissolving 1g of bromothymol blue in 50 cm^3 of distilled water. Mix well.

To make the 0.02% bromothymol blue solution, add 1cm^3 of this 2.0% stock solution to a beaker, make up to 100 cm^3 with distilled water and mix well.

A, at least 20cm^3 of 0.01 mol dm^{-3} sodium hydroxide solution in a beaker or container, labelled **A**.

This is prepared by putting 0.2g of solid sodium hydroxide (approximately one pellet) in 500 cm^3 of distilled water and mixing well.

Y, at least 15 cm^3 of 7 % yeast suspension, in a beaker or container, labelled **Y**. This volume should not include any froth.

Y should be prepared one hour before the examination. In a large container add 7.0 g of dried yeast (for baking) to 40 cm^3 of warm distilled water. Stir and make up to 100 cm^3 with distilled water. This should be kept for 30 to 40 minutes at a temperature of 35 °C to 40 °C.

10 to 15 minutes before the candidate starts **Question 1**, sprinkle 20g of glucose over the surface of the suspension and stir thoroughly.

SA, at least 15 cm^3 of 2 % sodium alginate solution in a small beaker or container, labelled **SA**.

This is prepared by dissolving 2g of sodium alginate in 50cm^3 of warm distilled water, stirring well and making up to 100cm^3 with warm distilled water. Continue stirring until dissolved. Cool to room temperature.

This should be prepared the day before the examination and kept refrigerated.

C, at least 50 cm^3 of 1.5 % calcium chloride solution in a beaker or container, labelled **C**.

This is prepared by dissolving 1.5 g of calcium chloride in 50 cm^3 of distilled water and making up to 100cm^3 with distilled water.

G, at least 40 cm^3 of 2 % glucose solution, in a beaker or container, labelled **G**. This is prepared by dissolving 2.0g of glucose in 50cm^3 of distilled water and making up to 100 cm^3 with distilled water.

It is advisable to wear safety glasses/goggles when handling chemicals.

Apparatus:

. **(i)** Four 10 cm^3 syringes or one with the means to wash it out. □

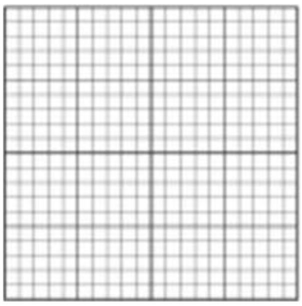
. **(ii)** Two 5 cm^3 syringes or one with the means to wash it out. □

. **(iii)** Container with tap water, labelled **For washing**. □

. **(iv)** Container, labelled **Waste**. □

. **(v)** One small beaker or container of maximum volume 25 cm^3. □

. **(vi)** Grid on white card 4cm × 4cm with 2mm grid lines. □



. **(vii)** One large test-tube. □

. **(viii)** 5 small test-tubes. □

. **(ix)** Bung(s) to fit small test-tubes. □

. **(x)** Test-tube rack or container to hold at least 6 test-tubes. □

. **(xi)** Glass rod. □

. **(xii)** Teat pipette. □

. **(xiii)** Petri dish or shallow container. □

. **(xiv)** Blunt forceps/ spatula. □

. **(xv)** Stop clock, stop watch or sight of a clock with a second hand. □

. **(xvi)** Glass marker pen. □

. **(xvii)** Safety goggles/glasses. □

Question 2

Solutions provided to the candidates should be supplied in a suitable beaker, or container, for removal of the solutions using a teat pipette. More of the solutions and materials should be available if requested by candidates.

. All solutions and materials should be provided to candidates at **room temperature**.

. Clean microscope slides, coverslips, teat pipettes and stage micrometer are needed for each candidate.

. Fresh **I**, **M**, **DW**, **S1 and slide S2** are needed for each candidate.

. Summary of solutions:

labelled	**contents**	**hazard**	**volume / cm^3**
I	iodine in potassium iodide solution	[H] irritant	at least 25
M	0.01% methylene blue solution	stains	at least 25
DW	distilled water	none	at least 25

Materials:

labelled	**contents**	**quantity**
S1	onion pieces	at least three pieces of onion wrapped in a damp paper towel, in a dish. These can be prepared the day before.

I, iodine in potassium iodide, at least 25 cm^3 in a bottle or container, with a pipette (teat), labelled **I**.

This is the same concentration as used when testing for starch. This is sufficient for one candidate.

M, at least 25 cm^3 of 0.01% methylene blue solution in a bottle or container, with a pipette (teat), labelled **M**.

This is sufficient for one candidate. **DW**, at least 25cm^3 of distilled water in a bottle or container, with a pipette (teat), labelled **DW**. This is sufficient for one candidate.

3 pieces of onion covered with damp paper towel in a dish, labelled **S1**. The three pieces are cut from an onion as shown in Fig. 2.1.

This is sufficient for one candidate.

Preparation of pieces of onion:

Candidates should not be given red onion. Onions with white flesh should be used, either with dry brown scales (yellow onion) or with dry white scales (white onion).

The onions should be as fresh as possible to avoid the effects of storage.

Cut off the top and bottom of the onion. Remove the outer dry scales. Cut the onion into pieces as in Fig. 2.1.

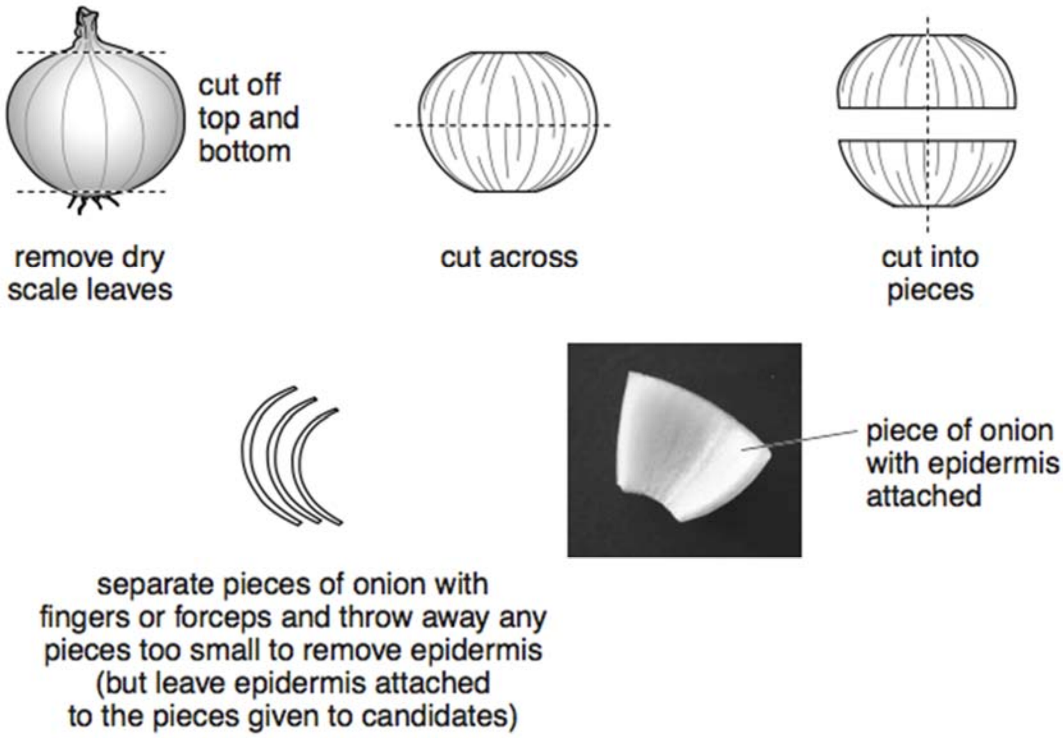


Fig. 2.1

(You are reminded to prepare spare pieces of onion to supply to the candidates as requested.)

The pieces of onion may be prepared before the examination and left overnight, at room temperature, in large containers with enough water to submerge the onion pieces. Cover the containers.

Apparatus for each candidate	Quantity
Clean microscope slides	3
Glass coverslips, e.g. 2 cm × 1 cm	3
Glass marker pen	1
Mounted needle or seeker or other means to position coverslip on microscope slide	1
White tile	1
Pipette, teat	3
Forceps	1
Sharp blade or scalpel	1
Container, to hold about 250 cm^3, labelled **For waste**	1
Paper towels	8
Microscope with: • Low-power objective lens, ×10 • High-power objective lens, ×40 • Eyepiece lens, ×10 • Eyepiece graticule fitted within the eyepiece and visible in focus at the same time as the specimen. For each candidate the microscope must be set up on low power.	1
Slide of *Ammophila sp.*, labelled S2	1
Stage micrometer	1